

Swaham Pitambar Patra

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import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# =====
# 1. LOAD DATASET
# =====
df = pd.read_csv('ECOMMERS SALES.csv')
print(f"Shape: {df.shape}")

# =====
# 2. CLEAN DATA
# =====
len_df = len(df)
missing_values = df.isnull().sum().sum()

print(f"Total Missing Values Found = {missing_values}")

if missing_values > 0:
    df = df.dropna()
    print(f"☑ Missing Values Cleared! Rows removed = {len_df - len(df)}")
else:
    print(f"☑ No missing values found. Dataset is already clean.")

print(f"Dataset shape after cleaning: {df.shape}")
print("-" * 50)

# =====
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# 3. CALCULATIONS FOR REPORT
# =====

total_sales = df["Total_Sales"].sum()

best_product =
df.groupby('Product_Name')['Quantity'].sum().idxmax()

highest_rev_category =
df.groupby('Category')['Total_Sales'].sum().idxmax()

avg_rev_per_category =
df.groupby('Category')['Total_Sales'].mean().round(2)

best_customer_segment =
df.groupby('Customer_Type')['Quantity'].sum().idxmax()

# =====
# 4. PRINT REPORT
# =====

print("\n" + "="*65)
print("----- E-COMMERCE SALES ANALYSIS REPORT -----")
print("="*65)
print(f"Total Sales = {total_sales}")
print(f"Best Selling Product = {best_product}")
print(f"Highest Revenue Category =
{highest_rev_category}")
print(f"Average revenue from each category
=\n{avg_rev_per_category}")
print(f"Customer Segment with Highest Purchases =
{best_customer_segment}")
print("="*65)

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# =====  
# 5. PLOTTINGS  
# =====  
  
# 1. Sales Trend Line Chart - Region, Total Sales  
plt.figure(figsize=(10, 6))  
region_sales =  
df.groupby('Region')['Total_Sales'].sum().sort_values(ascending=False)  
plt.plot(region_sales.index, region_sales.values,  
color='skyblue')  
plt.title('Sales by Region (Total Sales)')  
plt.xlabel('Region')  
plt.ylabel('Total Sales')  
plt.xticks(rotation=45)  
  
plt.show()  
  
# 2. Product Sales Bar Chart - Category, Total Sales  
plt.figure(figsize=(10, 6))  
category_sales =  
df.groupby('Category')['Total_Sales'].sum().sort_values(ascending=False)  
sns.barplot(x=category_sales.index,  
y=category_sales.values)  
plt.title('Product Sales by Category')  
plt.xlabel('Category')  
plt.ylabel('Total Sales')  
plt.xticks(rotation=45)  
  
plt.show()
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# 3. Revenue Heatmap
plt.figure(figsize=(12, 7))
sns.heatmap(
    df[["Total_Sales", "Quantity"]].corr(),
    annot = True
)
plt.title('Revenue Heatmap by Category and Region')

plt.show()

# 4. Customer Segmentation Chart - Pie Chart
plt.figure(figsize=(8, 8))
customer_sales =
df.groupby('Customer_Type')['Total_Sales'].sum()

plt.pie(customer_sales.values,
        labels=customer_sales.index,
        colors=["yellow", "blue", "red", "green"],
        )

plt.title('Customer Segmentation by Revenue',
fontsize=14)
plt.legend(customer_sales.index, title="Customer Type",
loc="best")
plt.axis('equal')
plt.show()

# 5. Profit Distribution Boxplot
plt.figure(figsize=(10, 6))
sns.boxplot(x='Category', y='Total_Sales', data=df)
plt.title('Sales Distribution by Category (Boxplot)')
plt.xlabel('Category')

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plt.ylabel('Total Sales')
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plt.show()
```

Output :

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 06-07> python .\E-Commerce_Sales_Analysis.py
Shape: (499, 10)
Total Missing Values Found = 0
✅ No missing values found. Dataset is already clean.
Dataset shape after cleaning: (499, 10)
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----- E-COMMERCE SALES ANALYSIS REPORT -----
=====

Total Sales = 9445124
Best Selling Product = Watch
Highest Revenue Category = Electronics
Average revenue from each category =
Category
Books          1618.02
Electronics    35270.08
Fashion        4799.67
Other          12283.49
Name: Total_Sales, dtype: float64
Customer Segment with Highest Purchases = New
=====
```

Plottings :





